Technical Report: Hybrid State-Dependent Ordering for Distributed Flatness
Certified Inversion Regularity and Switched ISS Tracking

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Abstract

We document a verification-first extension of a baseline distributed flatness result that assumes a fixed subsystem ordering. We introduce a state-dependent ordering policy and treat the resulting closed loop as a hybrid system. The report is structured as a pipeline of checkable assumptions and certified constants on a reachable trimmed set $\mathcal{R}$: (i) ordering/graph compatibility and hybrid well-posedness, (ii) mode-wise flatness via a permuted lower-triangular coupling structure, (iii) inversion regularity and conditioning via uniform pivot bounds for the baseline implicit-function inverses, and (iv) switched input-to-error stability in an ISS framework with explicit dwell-time and jump bounds. We provide an implementation plan and verification gates (Gates 1–4) that produce a constants table $(\underline{d}, \overline{c}, L_\Psi, J, \alpha, c_1, c_2, \mu, \nu, \tau_d)$ and determine whether a publishable theorem path is viable.

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1 Executive Summary and Deliverables

1.1 Objective

Given a baseline paper that provides a coupled flatness recursion under fixed ordering, our goal is to:

- formulate state-dependent ordering as a hybrid system with hysteresis/lockout,
- prove mode-wise flatness and define mode-dependent inversion maps Ψ_π via the baseline recursion,
- certify inversion conditioning and switching discontinuities on a reachable trimmed set $\mathcal{R}$,
- prove switched ISS tracking with an explicit dwell-time/jump inequality (and optionally eliminate jumps via blending),
- produce a reproducible verification repository and a constants table supporting auditability.

1.2 Outputs

- Technical report (this document) + appendices with proofs and verification methodology.
- Verification repository with scripts for Gates 1–4 producing JSON/CSV artifacts and plots.
- "Constants table" used directly in the final paper.

2 Baseline Setting and Recursion Objects

2.1 Subsystem structure and implicit inverses h_k^i

Each subsystem $i \in \{1, \dots, N\}$ is assumed to be in a pure-feedback / strict-feedback-like layered form of relative degree r. The baseline paper assumes nonsingularity of layer Jacobians enabling implicit inversion:

- For $k = 1, \dots, r-1$, the mapping $\bar{f}_k^i(\cdot)$ is nonsingular w.r.t. x_{k+1}^i, i.e. $D_{x_{k+1}^i} \bar{f}_k^i$ invertible.
- For $k = r$, the mapping $\bar{f}_r^i(\cdot)$ is nonsingular w.r.t. u^i, i.e. $D_{u^i} \bar{f}_r^i$ invertible.

By the implicit function theorem, there exist smooth inverse maps h_k^i such that (baseline Eq. (9)):

$$h_k^i(x_1^i, \ldots, x_k^i;\ \bar{f}_k^i(x_1^i, \ldots, x_{k+1}^i)) = x_{k+1}^i, \quad k = 1, \ldots, r-1, \tag{1}$$

$$h_r^i(x_1^i, \ldots, x_r^i;\ \bar{f}_r^i(x_1^i, \ldots, x_r^i, u^i)) = u^i. \tag{2}$$

2.2 Coupled recursion for Φ_k^i and reconstruction Ψ

Let $\zeta_{k-1} := (y, \dot{y}, \ldots, y^{(k-1)})$ denote stacked derivatives of the candidate flat output y. The baseline recursion (Eq. (10)–(11)) defines Φ_k^i:

$$\Phi_1^i(\zeta_0) = y^i, \tag{3}$$

$$\Phi_k^i(\zeta_{k-1}) = h_{k-1}^i\big(\Phi_1^i, \ldots, \Phi_{k-1}^i;\ D\Phi_{k-1}^i(\zeta_{k-1}) - \Delta_{k-1}^i(\Phi_1, \ldots, \Phi_{k-1}, \Phi_k^1, \ldots, \Phi_k^{i-1})\big), \tag{4}$$

for $k = 2, \ldots, r+1$, where $\Phi_k := (\Phi_k^1, \ldots, \Phi_k^N)$. The reconstructed input is:

$$u^i = \Phi_{r+1}^i(\zeta_r), \qquad \Psi(\zeta_r) := \Phi_{r+1}(\zeta_r). \tag{5}$$

Fixed ordering enters through the triangular dependence restriction: Δ_{k-1}^i may depend on current-layer terms Φ_k^j only for $j < i$.

3 Hybrid Formulation and Certified Domain

3.1 Ordering metric, modes, and guards

Choose an ordering metric $s_i(x)$ (PROJECT-SPECIFIC: define s_i explicitly, e.g. altitude). Define permutation $\pi(x)$ by deterministic sorting with a tie-break rule. Define mode regions and guard sets:

$$X_\pi = \{x :\ s_{\pi(1)}(x) > \cdots > s_{\pi(N)}(x)\}, \tag{6}$$

$$S_{ij} = \{x :\ s_i(x) = s_j(x)\}. \tag{7}$$

3.2 Switching logic with hysteresis and lockout

We implement mode memory with hysteresis margin $\varepsilon > 0$ and lockout dwell time τ_d:

- Maintain current mode π unless a candidate new permutation is stable by margin ε.
- Enforce lockout: once switched, disallow further switches for τ_d seconds (or N_d samples).

Define the trimmed domain away from ties:

$$X^\varepsilon := \bigcup_\pi X_\pi^\varepsilon, \qquad X_\pi^\varepsilon := \left\{x \in X_\pi :\ \min_{i \neq j} |s_i(x) - s_j(x)| \geq \varepsilon\right\}. \tag{8}$$

3.3 Reachable trimmed set $\mathcal{R}$ (operational definition)

We define a compact set $\mathcal{R} \subset X^\varepsilon$ on which all certificates and theorems apply. Operationally, we approximate:

$$\mathcal{R} \approx \left(\bigcup_{\xi \in \mathcal{X}_0} \mathrm{traj}(\xi)\right) \cap X^\varepsilon \cap \{x : \|x\| \leq R\}, \tag{9}$$

where $\mathrm{traj}(\xi)$ is the closed-loop trajectory from initial condition ξ under our hybrid controller. **Implementation:** we compute an empirical envelope via Monte Carlo simulations from $\mathcal{X}_0$, then shrink by a safety margin to define a conservative $\mathcal{R}$.

3.4 Checkable assumptions (boxed)

Assumption 1 (S1 Ordering regularity)**.** *Each $s_i \in C^1$. There exist $\varepsilon_0 > 0$, $m_s > 0$ such that if $|s_i(x) - s_j(x)| \le \varepsilon_0$ then $\|\nabla(s_i - s_j)(x)\| \ge m_s$.*

Assumption 2 (S2 Guard crossing transversality OR enforced lockout)**.** *Either (i) for guard crossings, $|\frac{d}{dt}(s_i - s_j)| \ge m_t > 0$, or (ii) lockout dwell time τ_d is enforced in the sampled implementation to prevent arbitrarily fast switching.*

Assumption 3 (G1 Metric–edge monotonicity)**.** *There exists $\gamma > 0$ such that on $\mathcal{R}$, for every active coupling edge $(i \to j) \in E(x)$, $s_i(x) \ge s_j(x) + \gamma$.*

Assumption 4 (P1′ Uniform pivot bound on $\mathcal{R}$)**.** *There exists $\underline{d} > 0$ such that on $\mathcal{R}$: for all i and $k \le r$, $\sigma_{\min}(D_{x_k^i} \bar{f}_{k-1}^i) \ge \underline{d}$, and for $k = r+1$, $\sigma_{\min}(D_{u^i} \bar{f}_r^i) \ge \underline{d}$.*

Assumption 5 (A2 Uniform derivative bounds)**.** *There exists $\bar{c}$ such that on $\mathcal{R}$ (and induced ζ-sets), all required partial derivatives of $\bar{f}_k^i$ and Δ_k^i are bounded by $\bar{c}$, and Δ_k^i is Lipschitz in its arguments with constant $\le \bar{c}$.*

Assumption 6 (F1 Mode-invariant output regularity (optional))**.** *If we claim a common labeled flat output y, then the required decoupling/rank conditions for that y hold on $\mathcal{R}$ for all modes. Otherwise we use mode-dependent outputs y_π.*

4 Gate 1: Graph Compatibility and Hybrid Well-Posedness

4.1 What Gate 1 computes

Gate 1 evaluates along trajectories in $\mathcal{R}$:

- Build the active coupling graph $G(x)$ using a theory-consistent edge rule (PROJECT-SPECIFIC).
- Check acyclicity and whether $\pi(x)$ is a topological ordering.
- Record switching statistics (min inter-switch time/samples) under hysteresis+lockout.

4.2 Pass/fail criteria

- PASS if cycle rate is zero on $\mathcal{R}$ and topological order pass rate is 1.0.
- FAIL if cycles occur: either redefine s_i, redefine edges, or restrict $\mathcal{R}$.
- PASS if lockout is respected and no sample-to-sample chattering occurs.

5 Mode-wise Flatness Under State-Dependent Ordering

Lemma 1 (Triangular dependence after ordering)**.** *Under Assumption G1, in each mode π the coupling dependencies in Δ_{k-1}^i respect the topological ordering induced by π, so the recursion remains forward-substitutable after permutation.*

Walkthrough (documentation-level): G1 implies every active edge points from higher metric to lower metric with margin γ. Therefore the active graph is acyclic and admits the sorted order as a topological order. Replacing the baseline fixed index order by π simply reindexes "earlier" vs "later" subsystems. Thus, for each i in mode π, Δ_{k-1}^i depends on current-layer terms Φ_k^j only for those j preceding i in π.

Theorem 1 (Mode-wise flatness)**.** *Fix a mode π and restrict to $\mathcal{R} \cap X_\pi^\varepsilon$. Under the baseline subsystem assumptions, Lemma 1, and Assumptions P1′, A2, the recursion defines smooth maps Φ_k^i and reconstructs (x, u) from $(y, \dot{y}, \dots)$. If Assumption F1 holds, the same labeled output y is a flat output for all modes; otherwise the system is piecewise flat with mode-dependent outputs y_π.*

6 Certified Inversion Regularity and Switching Discontinuities (Gate 3)

6.1 Lemma 3: Regularity of Φ and Lipschitz bound for Ψ

Lemma 2 (Uniform regularity and Lipschitz bound)**.** *Fix a mode π and restrict to $\mathcal{R} \cap X_\pi^\varepsilon$. Under Assumptions P1′, A2, and triangular dependence, for each k the map Φ_k is well-defined and C^1 on the induced compact set of ζ_{k-1}. Moreover, there exists L_Ψ such that for all $\zeta, \zeta' \in \mathcal{Z}_r$,*

$$\|\Psi_\pi(\zeta) - \Psi_\pi(\zeta')\| \le L_\Psi \|\zeta - \zeta'\|, \quad L_\Psi \le \frac{K(\bar{c}, N, r)}{\underline{d}^p}.$$

Walkthrough (how the constants arise):

1. (IFT step) Each h_{k-1}^i solves for x_k^i (or u^i at the last step). By IFT, its Lipschitz constant can be bounded by the inverse of the pivot Jacobian times a bound on remaining partials:

$$L_{h,k-1} \lesssim \|(D_{x_k^i} \bar{f}_{k-1}^i)^{-1}\| \cdot (\text{bounded partials}) \le c_h(\bar{c})/\underline{d}.$$

2. (Composition) Φ_k^i is a composition of h_{k-1}^i with arguments depending on earlier Φ maps and Δ. Use triangle inequality and Lipschitz bounds of Δ (bounded by $\bar{c}$) to obtain a recurrence $L_k \le L_{h,k-1}(\cdots)$.

3. (Triangular structure) Because Δ_{k-1}^i depends on Φ_k^j only for earlier j, we can solve $i = 1 \to N$ and stack the bounds. The recursion depth and repeated inversions yield the exponent p in $\underline{d}^{-p}$.

6.2 Lemma 3J: Jump bound and optional blending

Lemma 3 (Switch jump bound)**.** *Let $J := \sup_{t_s} \|u(t_s^+) - u(t_s^-)\|$ over switch times t_s with $x(t_s) \in \mathcal{R}$. Then*

$$J \le \sup_{\zeta \in \mathcal{Z}_r} \|\Psi_{\pi^+}(\zeta) - \Psi_{\pi^-}(\zeta)\|.$$

If input blending is enabled in a transition band, then $J = 0$ and $\|\dot{u}\|$ is bounded by an explicit expression involving $\|w'\|_\infty$, $\sup|\dot{\rho}|$, and $\sup\|u_\pi - u_{\pi'}\|$ on $\mathcal{R}$.

6.3 Constants table (must be populated by Gate 3)

7 Switched/Hybrid Tracking Stability (Gate 4)

7.1 Controller and disturbance definitions

Let $\pi(t)$ be the mode signal generated by hysteresis+lockout. The implemented control is

$$u(t) = \Psi_{\pi(t)}(\zeta_r(t)).$$

Define a reference $u^\star(t)$ (Case A: any consistent mode; Case B: mode-matched ideal inverse) and mismatch $\Delta u(t) = u(t) - u^\star(t)$.

Constant	Definition	How computed/certified
$\underline{d}$	min singular value of pivot Jacobians	sampled min over $\mathcal{R}$
$\overline{c}$	max derivative/Lipschitz bounds	sampled max (or analytic bound)
L_Ψ	Lipschitz constant bound for Ψ	Lemma 3 recurrence + $\underline{d}, \overline{c}$
J	switch jump bound	measured from switch events (or 0 with blending)

Table 1: Core certified constants produced by Gate 3.

Theorem 2 (Hybrid ISS tracking with dwell time)**.** *Assume for each mode π there exists a Lyapunov function $V_\pi(e)$ and constants $c_1, c_2, \alpha, \beta > 0$ such that*

$$c_1\|e\|^2 \le V_\pi(e) \le c_2\|e\|^2, \qquad \dot{V}_\pi \le -\alpha\|e\|^2 + \beta\|\Delta u\|^2.$$

Assume that at switches t_s,

$$V_{\pi^+}(e(t_s^+)) \le \mu V_{\pi^-}(e(t_s^-)) + \nu\|u(t_s^+) - u(t_s^-)\|^2$$

for $\mu \ge 1$, $\nu \ge 0$, and that a dwell time τ_d is enforced. If

$$\mu \exp\left(-\frac{\alpha}{c_2}\tau_d\right) < 1,$$

then the tracking error is ISS with respect to Δu and jump size J on $\mathcal{R}$. If blending enforces $J = 0$, the jump term vanishes.

7.2 Gate 4 verification objective

Gate 4 empirically validates the *trend* implied by the theorem: increasing τ_d and/or decreasing J (via blending) improves boundedness/decay of $\|e(t)\|$ under noise and delay.

8 Repository and Script Interface (Implementation Mapping)

8.1 Math-to-code mapping (required)

Math object	Code location
$s_i(x)$, $\pi(x)$, hysteresis/lockout	`src/hybrid/ordering.py`
$\Delta_k^i(\cdot)$ and edge rule $E(x)$	`src/model/coupling.py`, `src/model/graph.py`
Φ_k^i recursion (Eq. 10)	`src/flatness/recursion.py`
Pivot Jacobians and $\underline{d}$	`src/flatness/pivots.py`
Blending $w(\rho)$ (optional)	`src/hybrid/blending.py`
Gate 1–4 scripts	`src/verify/gate*.py`

Table 2: Mapping between paper objects and implementation.

A Appendix: Verification Gates (Pass/Fail)

This appendix should mirror the repository outputs: Gate 1 (graph + switching), Gate 2 (flat output rank), Gate 3 (constants), Gate 3.5 (satisfiability), Gate 4 (Monte Carlo trend).